# CY202 — Complete Answer Set

Every topic in the Gemini topic list, answered: short 2-mark questions first, then the 7-mark questions, with the diagrams that carry marks and nothing repeated

| | | | |
|---|---|---|---|
| **19**<br>two-mark answers | **27**<br>seven-mark answers | **18**<br>diagrams, each used once | **6**<br>numerical types |

## How this is organised

**Part A — Short answers (2 marks).** Definitions and one-line facts, written the way they are marked: definition, then two or three qualifying points. These are the marks that are won or lost in a minute each.

**Part B — Long answers (7 marks).** Each answer follows the same skeleton — one-line definition, then numbered or bulleted points, then significance — so the shape is the same every time and only the content changes.

**Part C — Numericals.** The six formula types the briefing says to expect, each with one worked line.

**Diagrams.** Included only where the answer needs one, and used once: the geosequestration diagram does double duty for carbon sequestration and CCS, so it appears only under that answer.

Answers are deliberately point-to-point rather than continuous prose, because the briefing asks for answers that meet the merit of the question and not five-page essays. Bold text marks the term the examiner looks for.

# PART A
# Short-answer questions — 2 marks

Definitions and one-line facts. Aim for four to six lines, with the key term written exactly as the examiner expects it.

## S1. What is a peptide bond?

**2 marks**

A covalent **amide bond (–CO–NH–)** formed between the α-carboxyl group of one amino acid and the α-amino group of the next, with the loss of one water molecule (condensation).

- It is **planar and rigid** — the C–N bond has partial double-bond character, so rotation is restricted.
- It is formed on the ribosome during translation and is broken by hydrolysis (proteases).
- Peptide bonds link amino acids into di-, tri-, oligo- and polypeptides (proteins).

## S2. What is a glycosidic linkage?

**2 marks**

A covalent bond formed by **condensation (loss of $H_2O$)** between the anomeric carbon of one monosaccharide and a hydroxyl (–OH) group of another sugar.

- α-1,4 in starch and glycogen; β-1,4 in cellulose; α-1,2 in sucrose.
- It joins monosaccharides into disaccharides, oligosaccharides and polysaccharides, and is broken by hydrolysis (amylase, cellulase).

## S3. Define dipeptide.

**2 marks**

A molecule made of **two amino acids joined by one peptide bond** (e.g. glycyl-alanine). On hydrolysis it gives exactly two amino acids.

## S4. Define tripeptide.

**2 marks**

A molecule made of **three amino acids joined by two peptide bonds** (e.g. glutathione: γ-glutamyl-cysteinyl-glycine, an important cellular antioxidant).

## S5. What are oligosaccharides?

**2 marks**

Carbohydrates containing **2 to 10 monosaccharide units** joined by glycosidic bonds — the term covers disaccharides (sucrose, lactose, maltose) and short chains such as raffinose and stachyose.

- Water-soluble, sweet, and often attached to proteins and lipids to form glycoproteins and glycolipids that act in cell recognition.
- On hydrolysis they yield their constituent monosaccharides.

## S6. What are transmembrane proteins? Give their significance and functions.

**2 marks**

**Integral membrane proteins that span the whole lipid bilayer** — single-pass (e.g. glycophorin) or multi-pass (e.g. bacteriorhodopsin, GPCRs). Their transmembrane segments are hydrophobic (α-helices of non-polar residues), while the ends projecting into the aqueous phase are hydrophilic.

- **Functions:** channels and pores (aquaporins); carriers/pumps ($Na^+$–$K^+$ ATPase, GLUT transporters); receptors and signal transduction (G-protein coupled receptors); cell adhesion and anchoring; identity antigens (blood-group glycoproteins).
- **Significance:** they control selective permeability, ~30% of cell proteins are membrane proteins, and they are the targets of most modern drugs.

## S7. Name the disease caused by deficiency of Vitamin A.

**2 marks**

**Night blindness (nyctalopia)**, progressing to **xerophthalmia** — dry conjunctiva with Bitot's spots and, in severe cases, **keratomalacia** (corneal softening and blindness). Growth retardation and lowered immunity also occur.

## S8. Name the disease caused by deficiency of Vitamin B9 (folic acid).

**2 marks**

**Megaloblastic anaemia** — large, immature red blood cells. In pregnancy it causes **neural tube defects** in the foetus (spina bifida), and it also causes fatigue, irritability and poor growth.

## S9. Name the metal centre present in chlorophyll.

**2 marks**

**Magnesium — $Mg^{2+}$**, held at the centre of the porphyrin (chlorin) ring of the chlorophyll molecule. It is essential for absorption of light and for electron transfer in photosystems I and II; without Mg the pigment loses its function.

## S10. Name the metal centre present in Vitamin B12.

**2 marks**

**Cobalt — Co** (as $Co^{3+}$), at the centre of the corrin ring of cobalamin. It is needed for red blood cell formation and nerve function; deficiency causes pernicious (megaloblastic) anaemia.

## S11. Name the metal centre present in Vitamin D3 / carboxylase, with its oxidation state.

**2 marks**

**Calcium — $Ca^{2+}$, oxidation state +2.** $Ca^{2+}$ is the cofactor of carboxylase enzymes (e.g. the vitamin-K-dependent clotting factors and osteocalcin), where it bridges the enzyme to its substrate.

## S12. Write any four advantages of bio-fertilizers.

**2 marks**

- **Renewable and cheap** — prepared from locally available cultures and organic matter.
- **Eco-friendly:** no chemical pollution of soil, water or produce; safe for humans.
- **Nitrogen fixation** by *Rhizobium*, *Azotobacter*, *Azospirillum* and blue-green algae; **phosphate solubilisation** by *Pseudomonas* and *Bacillus*.
- **Improves soil health** — adds organic matter, improves structure, water-holding capacity and microbial activity, and raises yield by about 10–20% while cutting chemical fertiliser use.

## S13. Write any four advantages of bio-fuels.

**2 marks**

- **Renewable** and produced from agricultural, forest and municipal waste — energy from waste.
- **Nearly carbon-neutral:** the $CO_2$ released was absorbed by the plant during growth; much lower $SO_2$ and particulate emissions than fossil fuels.
- **Biodegradable, non-toxic** and safer to handle than petroleum fuels.
- **Reduces import dependence** and supports rural employment and farmers' income (ethanol from sugarcane, biodiesel from jatropha).

## S14. Define Noise Standards, 2000 and give the ambient noise limit values.

**2 marks**

The **Noise Pollution (Regulation and Control) Rules, 2000**, framed under the Environment (Protection) Act 1986, fix ambient noise limits for different zones. Limits are in dB(A), as Leq, day (6 am–10 pm) / night (10 pm–6 am):

- **Industrial zone:** 75 / 70
- **Commercial zone:** 65 / 55
- **Residential zone:** 55 / 45
- **Silence zone** (100 m around hospitals, schools, courts): **50 / 40**

## S15. What is a food chain?

**2 marks**

The **linear sequence of organisms through which energy and nutrients pass**, from the producer, through herbivores, to carnivores, ending with decomposers.

- **Grazing (predator) chain:** grass → grasshopper → frog → snake → hawk.
- **Detritus (decomposer) chain:** dead leaves → earthworm → bird.

## S16. What is a food web?

**2 marks**

The **network of interconnected food chains** in an ecosystem, in which one organism feeds on, or is eaten by, several others.

- It shows the real feeding relationships and gives the ecosystem **stability and resilience** — if one link fails, energy can flow through alternative pathways.

## S17. What is an environmental gradient?

**2 marks**

A **gradual, directional change in an environmental factor** — temperature, light, moisture, salinity, altitude or soil nutrients — across space or time.

- Species abundance and distribution change along it; each species survives only between its limits of tolerance, giving the **tolerance curve** (optimum, zones of stress, zones of intolerance).

## S18. Define GPP and NPP, and state their relation.

**2 marks**

**Gross primary productivity (GPP)** is the total organic matter fixed by producers by photosynthesis per unit area per unit time. **Net primary productivity (NPP)** is what remains after the producers' own respiration (R).

**NPP = GPP − R.** NPP is the energy actually available to herbivores, so it controls the yield of the whole ecosystem.

## S19. What are the different R's used in waste management?

**2 marks**

The classic **3 R's** are **Reduce** (prevent waste at source), **Reuse** (use again without reprocessing) and **Recycle** (convert back to raw material).

- Extended to the **9 R's** in the circular economy, in order of preference: **Refuse → Rethink → Reduce → Reuse → Repair → Refurbish → Remanufacture → Repurpose → Recycle**, with **Recover** (energy) and **Rot** (composting) added at the end.
- The order matters: the higher options save more material and energy, so they are preferred over recycling and disposal.

# PART B
# Long-answer questions — 7 marks

Definition, then the body in points, then significance. Where a diagram is shown, draw it — the label marks are inside the diagram.

## L1. Describe the fundamental characteristics of living organisms.

**7 marks**

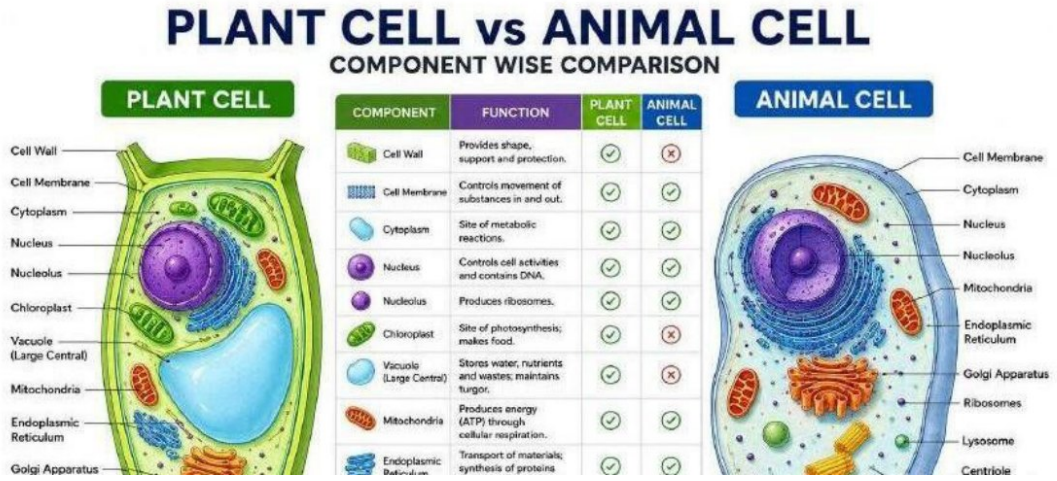
Living organisms are distinguished from non-living matter by a set of cardinal features; all of them are shown by a single cell, so the cell is the unit of life.

- **Cellular organisation** — protoplasm organised into one or more cells with organelles.
- **Metabolism** — continuous anabolism and catabolism (photosynthesis, respiration) with exchange of matter and energy.
- **Growth** — increase in size and mass by internal addition of protoplasm.
- **Reproduction** — ability to produce offspring, sexually or asexually, ensuring continuity of the species.
- **Responsiveness (irritability)** — sensitivity to stimuli such as light, heat, chemicals and touch.
- **Homeostasis** — regulation of internal conditions (temperature, pH, water, ions) within narrow limits.
- **Movement/locomotion** — internal streaming or movement of the whole organism.
- **Adaptation and evolution** — heritable change in response to environment over generations.
- **Definite life span** and death; plus **heredity**, the transmission of characters through genes.

## L2. Compare the structure of plant, animal and bacterial cells.

**7 marks**

A recurring examination and conceptual theme in cell biology is the systematic comparison of plant, animal, and bacterial cells. The following consolidated tables present these comparisons across structural, biochemical, and physiological criteria.




*Comparative structure of plant and animal cells. The bacterial cell differs in being prokaryotic — peptidoglycan wall, nucleoid instead of nucleus, 70S ribosomes, no membrane-bound organelles.*

Plant and animal cells are **eukaryotic** (true nucleus, membrane-bound organelles); the bacterial cell is **prokaryotic** (nucleoid, no membrane-bound organelles).

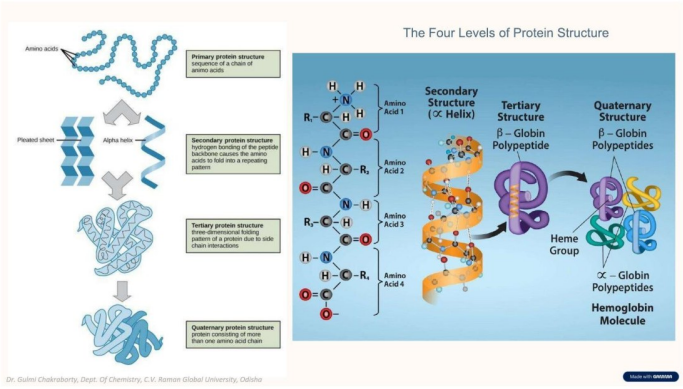
- **Cell wall:** present in plant (cellulose) and bacteria (peptidoglycan); absent in animal cells.

- **Nucleus:** true, membrane-bound in plant and animal; absent in bacteria (single circular DNA in the nucleoid).
- **Chloroplasts:** present in plant cells (photosynthesis); absent in animal and bacterial cells (some bacteria have pigment vesicles).
- **Vacuole:** one large central vacuole in plant cells; small or absent in animal cells; absent in bacteria.
- **Mitochondria, ER, Golgi, lysosomes:** present in plant and animal cells; absent in bacteria.
- **Ribosomes:** 80S in eukaryotes, 70S in bacteria.
- **Size:** plant/animal 10–100 µm; bacteria 1–10 µm. **Centrosome:** present in animal cells only. **Flagella:** simple in bacteria, complex in eukaryotes.

## L3. Distinguish between the primary, secondary, tertiary and quaternary structures of proteins, and give the significance of each.

**7 marks**

| Level | Description |
|---|---|
| Primary | The linear sequence of amino acids joined by covalent peptide bonds, read N-terminus to C-terminus; encoded directly by the gene. |
| Secondary | Local, regular folding patterns stabilised by hydrogen bonds between backbone -NH and -C=O groups: the α-helix (right-handed coil) and β-sheet (extended, laterally H-bonded strands). |
| Tertiary | The overall 3-D folding of the entire polypeptide chain, stabilised by hydrophobic interactions, hydrogen bonds, ionic bonds, disulphide bridges (-S-S-) and van der Waals forces; creates the active site. |
| Quaternary | The association of two or more polypeptide subunits into a single functional complex, e.g., haemoglobin (2α + 2β chains), stabilised by the same non-covalent forces plus inter-subunit disulphide bonds. |




*The four levels of protein structure, illustrated with haemoglobin.*

Protein structure is hierarchical — each level builds on the one below.

- **Primary structure** — the linear sequence of amino acids joined by peptide bonds. Significance: the sequence is coded by DNA and determines all higher levels; a single change can cause disease (sickle-cell anaemia).
- **Secondary structure** — regular local folding into α-helix and β-pleated sheet, stabilised by **hydrogen bonds** between backbone C=O and N–H groups. Significance: gives shape, elasticity and strength (keratin, silk fibroin).
- **Tertiary structure** — the overall three-dimensional shape of one chain, stabilised by hydrophobic interactions, hydrogen bonds, ionic bonds and **disulphide bridges**. Significance: creates the active site and hence the function of globular proteins (enzymes, antibodies).
- **Quaternary structure** — association of two or more polypeptide subunits, e.g. haemoglobin (four subunits). Significance: enables co-operativity and regulation of activity — the classic example of structure determining function.

## L4. Classify carbohydrates, giving examples.

**7 marks**

Carbohydrates are polyhydroxy aldehydes or ketones, classified by the number of sugar units.

- **Monosaccharides** (one unit, CnH2nOn): trioses (glyceraldehyde), pentoses (ribose, deoxyribose), hexoses (glucose, fructose, galactose). Classified as **aldoses** (glucose) or **ketoses** (fructose).
- **Disaccharides** (two units, glycosidic bond): sucrose (glucose + fructose, α-1,2), lactose (glucose + galactose, β-1,4), maltose (glucose + glucose, α-1,4).
- **Oligosaccharides** (2–10 units): raffinose, stachyose; found in glycoproteins and glycolipids.
- **Polysaccharides** (many units). *Homopolysaccharides:* starch (plant storage), glycogen (animal storage), cellulose (plant cell wall), chitin (fungal wall, insect exoskeleton). *Heteropolysaccharides:* hyaluronic acid, heparin, peptidoglycan.
- **Functions:** immediate and stored energy (glucose, starch, glycogen); structural support (cellulose, chitin); cell recognition (glycoproteins); precursors of nucleic acids (ribose, deoxyribose).

## L5. What are enzymes? Discuss their types and classification.

**7 marks**

Enzymes are **biological catalysts**, almost all proteins, that accelerate biochemical reactions by lowering activation energy without being consumed. They act through a specific active site (lock-and-key or induced-fit) and are sensitive to temperature and pH.

- **IUBMB classification — six classes:** 1 **Oxidoreductases** (dehydrogenases, oxidases, catalase) — oxidation–reduction; 2 **Transferases** (kinases, transaminases) — transfer of a group; 3 **Hydrolases** (amylase, lipase, protease) — hydrolysis; 4 **Lyases** (decarboxylase, aldolase) — removal of a group without hydrolysis; 5 **Isomerases** (isomerase, mutase) — intramolecular rearrangement; 6 **Ligases** (DNA ligase, synthetase) — joining with ATP.
- **By composition:** simple (only protein) and conjugated (apoenzyme + cofactor). Cofactors may be metal ions ($Mg^{2+}$, $Zn^{2+}$) or coenzymes (NAD, FAD, CoA, vitamins).
- **By site of action:** intracellular (most) and extracellular (digestive enzymes). Also classified as constitutive (always present) and inducible (formed on demand).
- **Inhibition:** competitive (substrate analogue at active site), non-competitive (allosteric site) and uncompetitive — the basis of many drugs and pesticides.

## L6. What are pesticides? Classify them and give their advantages and disadvantages.

**7 marks**

**Pesticides** are chemical or biological agents used to kill or control pests — insects, weeds, fungi, rodents, nematodes and molluscs — that damage crops or spread disease.

- **By target (8 types): insecticides** (DDT, malathion), **herbicides** (2,4-D, glyphosate), **fungicides** (Bordeaux mixture, captan), **rodenticides** (warfarin, zinc phosphide), **nematicides** (carbofuran), **molluscicides** (metaldehyde), **acaricides** (dicofol), **bactericides** (streptomycin).
- **By chemical class:** organochlorines (DDT, BHC — persistent, biomagnify), organophosphates (malathion, parathion — cholinesterase inhibitors), carbamates (carbaryl), pyrethroids (permethrin), and bio-pesticides (*Bt*, neem).
- **Advantages:** protect yield and stored grain; control disease vectors (malaria, plague); cheap, fast and easy to apply; raise farm income and food security.
- **Disadvantages: biomagnification** and persistence (DDT in food chains); **pest resistance** and resurgence; kill **non-target and beneficial** organisms (bees, predators); residues in food; soil and groundwater pollution; toxicity to humans (organophosphates); secondary pest

outbreaks.

## L7. What is biological pest control? Discuss its methods.

**7 marks**

**Biological pest control** is the use of living organisms — predators, parasitoids and pathogens — to suppress pest populations, instead of chemicals. It is species-specific, non-polluting and self-sustaining.

- **1. Classical (importation) control:** the pest's natural enemy is introduced from its native range and established permanently. Example: the cottony-cushion scale was controlled by the ladybird *Vedalia*; prickly pear by the moth *Cactoblastis*.
- **2. Augmentation:** mass-reared enemies are released periodically — *inundation* (large numbers, quick kill) or *inoculation* (small numbers, they multiply). Example: *Trichogramma* wasps against sugarcane borers.
- **3. Conservation:** farming practices that protect existing enemies — strip cropping, beetle banks, avoiding broad-spectrum sprays, providing nectar sources.
- **Agents used: predators** (ladybird vs aphids, spider, praying mantis), **parasitoids** (*Trichogramma*, *Encarsia*), **pathogens** — bacteria (*Bacillus thuringiensis*), viruses (NPV), fungi (*Trichoderma*, *Beauveria*), and botanical agents (neem, pyrethrum).
- **Limitations:** slow action, needs correct identification and monitoring, climate sensitivity, and risk to non-target species if poorly chosen.

## L8. What are vaccines? Explain the types of vaccines and how they work.

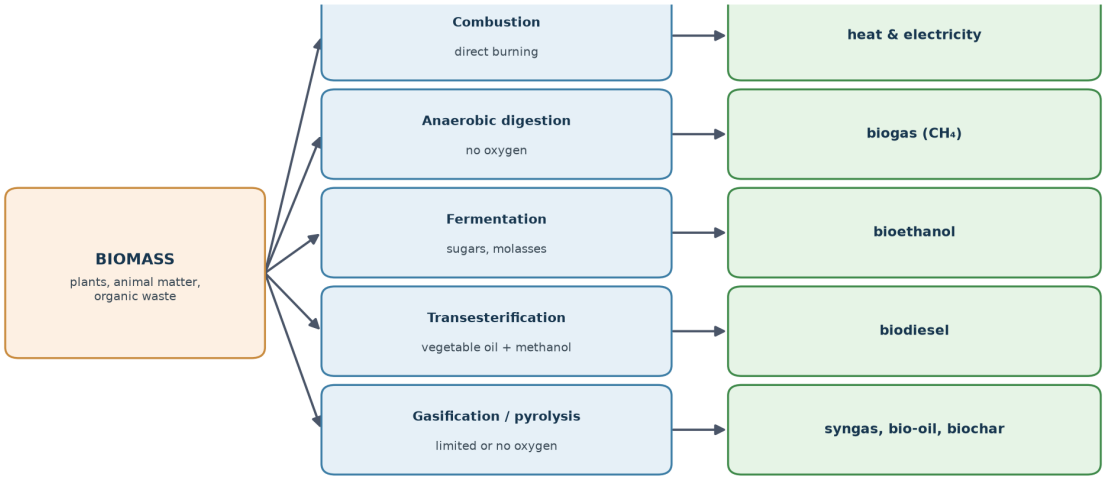
**7 marks**

A **vaccine** is a preparation of weakened, killed or fragmented antigen (or its genetic instructions) that stimulates the immune system to produce antibodies and memory cells without causing disease. On later exposure, memory B and T cells give a rapid, strong secondary response — so the person is immune.

- **Live attenuated:** weakened live pathogen — strong, long-lasting immunity. Examples: BCG, MMR, oral polio (Sabin), varicella. Caution in immunocompromised patients.
- **Inactivated (killed):** killed pathogen — safer, needs boosters. Examples: Salk polio, hepatitis A, pertussis.
- **Toxoid:** inactivated bacterial toxin. Examples: tetanus, diphtheria.
- **Subunit / conjugate / recombinant:** purified antigen only. Examples: hepatitis B (recombinant surface antigen), Hib and pneumococcal conjugate, HPV.
- **mRNA and viral-vector vaccines:** the cell is given the gene for the antigen and makes it itself. Examples: COVID-19 mRNA (Pfizer, Moderna), adenovirus vector (Covishield).
- **Importance:** prevents epidemics, eradicates disease (smallpox), gives herd immunity, and cuts mortality and health costs.

## L9. Define bio-energy. What are its types, advantages and disadvantages?

**7 marks**




*Bio-energy conversion routes*

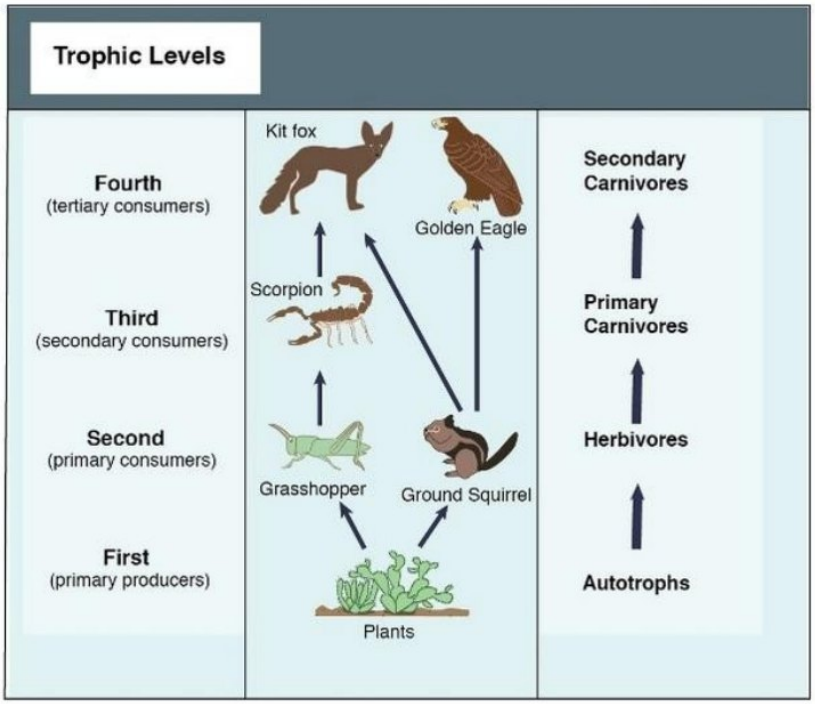
*Bio-energy conversion routes — biomass to heat, biogas, bioethanol, biodiesel and syngas.*

**Bio-energy** is renewable energy obtained from biomass — plant and animal matter, and organic waste — through combustion, digestion or conversion into fuels.

- **Types:** (i) **biomass combustion** — wood, crop residue, bagasse for heat and electricity; (ii) **biogas** — anaerobic digestion of dung and waste (60% $CH_4$) for cooking and power; (iii) **bioethanol** — fermentation of sugarcane, maize, molasses; (iv) **biodiesel** — transesterification of vegetable oil/jatropha with methanol; (v) **gasification and pyrolysis** — producer gas, bio-oil, biochar; (vi) **biohydrogen** from algae and dark fermentation.
- **Advantages:** renewable and carbon-neutral; made from waste (energy from waste); cheap and locally available; less $SO_2$ and particulates than coal; reduces landfill; supports rural income and energy security.
- **Disadvantages:** low energy density and land/water requirement; competition with food crops; deforestation if unmanaged; air pollution from inefficient burning; seasonal availability; technology and capital cost.

## L10. What are trophic levels? How does energy flow through them, and how do the laws of thermodynamics apply?

**7 marks**




*Trophic levels — energy passes from autotrophs up through herbivores to carnivores, losing about 90% at every step (the 10% law).*

A **trophic level** is the feeding position an organism occupies in a food chain. **T1** producers (plants), **T2** primary consumers/herbivores, **T3** secondary consumers, **T4** tertiary consumers, with decomposers acting at every level.

- Energy flow is **unidirectional**: sunlight → producers → consumers → decomposers, and is lost as heat at each step. It never cycles back.
- **10% law (Lindeman):** only about 10% of the energy at one level is transferred to the next; 90% is lost in respiration, as heat, in excretion and as uneaten parts. So biomass and numbers fall sharply at higher levels, giving the **pyramid of energy** (always upright).
- **First law of thermodynamics** (energy is conserved): the energy entering the ecosystem as sunlight equals that fixed by photosynthesis plus that reflected and lost as heat — nothing is created.
- **Second law of thermodynamics** (entropy increases): at every transfer some useful energy is degraded to heat, so energy transfer is never 100% efficient — this is the thermodynamic reason for the 10% law and for the short length of food chains (4–5 levels).
- **Consequence:** eating lower on the food chain (plants) feeds more people than eating meat, and this is why the number of trophic levels is limited.

## L11. Describe the nitrogen cycle and the processes involved.

**7 marks**

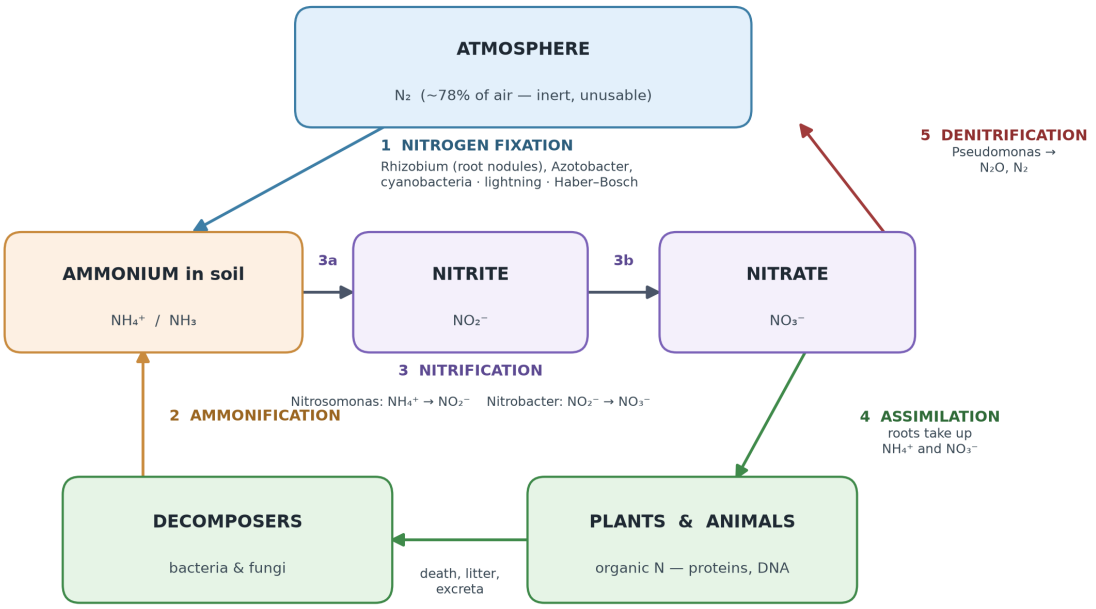



*Fig. 1 The nitrogen cycle — five processes: fixation (1), ammonification (2), nitrification (3), assimilation (4), denitrification (5).*

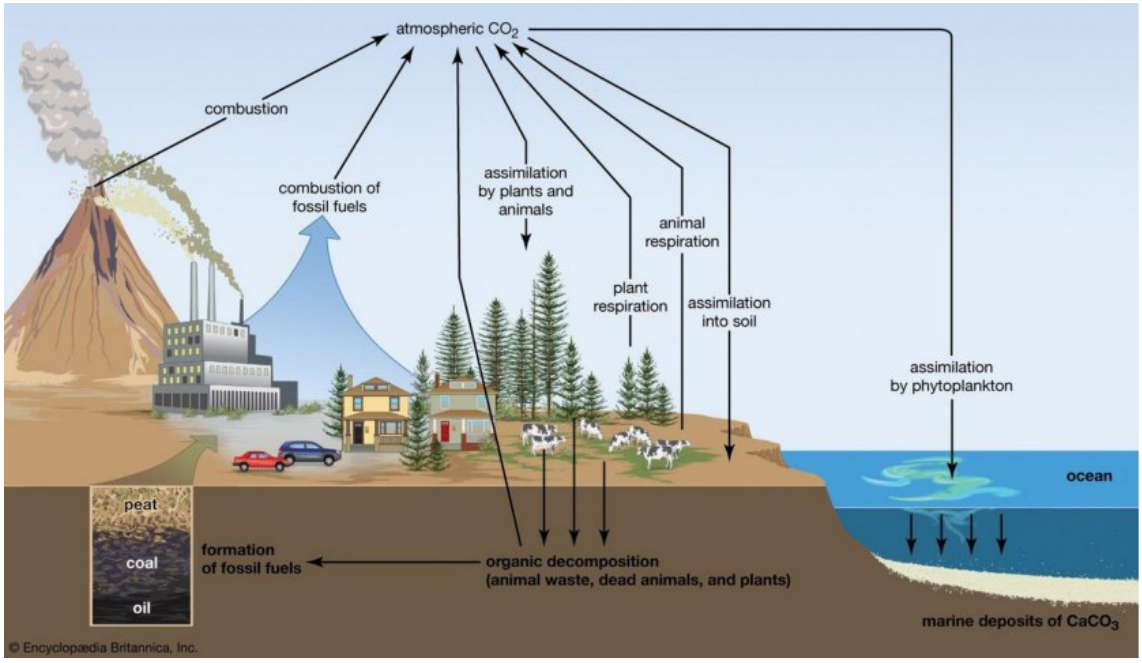
*The nitrogen cycle — fixation, ammonification, nitrification, assimilation and denitrification.*

The **nitrogen cycle** is the biogeochemical cycle by which nitrogen is converted among its chemical forms and circulated through atmosphere, soil, water and organisms. Air is 78% $N_2$, but the triple bond makes it inert, so usable nitrogen limits growth.

- **1. Nitrogen fixation** — $N_2 \rightarrow NH_3/NH_4^+$ by *Rhizobium* (root nodules), free-living *Azotobacter*, *Clostridium* and cyanobacteria (enzyme nitrogenase); also by lightning and by the Haber–Bosch process for fertiliser.
- **2. Ammonification** — decomposers (*Bacillus*, *Pseudomonas*, fungi) break down dead matter, urine and urea, releasing $NH_4^+$ into the soil.
- **3. Nitrification** — $2NH_4^+ + 3O_2 \rightarrow 2NO_2^- + 2H_2O + 4H^+$ (*Nitrosomonas*), then $2NO_2^- + O_2 \rightarrow 2NO_3^-$ (*Nitrobacter*). Nitrate is the form plants absorb most readily.
- **4. Assimilation** — plants take up $NH_4^+$ and $NO_3^-$ and reduce them into amino acids, proteins and nucleic acids; animals obtain nitrogen by feeding on plants or other animals.
- **5. Denitrification** — *Pseudomonas denitrificans* reduces $NO_3^- \rightarrow NO_2^- \rightarrow NO \rightarrow N_2O \rightarrow N_2$ under anaerobic conditions, returning nitrogen to the atmosphere and closing the cycle. Losses also occur by leaching of nitrate and volatilisation of ammonia.
- **Human impact:** fertiliser manufacture, fossil-fuel NOx and wastewater have roughly doubled fixation — causing acid deposition, eutrophication, nitrate in drinking water and $N_2O$ (a potent greenhouse gas).

## L12. Discuss the carbon cycle.

**7 marks**




*The carbon cycle — photosynthesis, respiration, decomposition, ocean exchange and fossil-fuel combustion.*

The **carbon cycle** is the biogeochemical cycle by which carbon moves between the atmosphere, oceans, soil, living organisms and rocks. Carbon is the backbone of all organic molecules, so the cycle governs life and climate.

- **Reservoirs:** atmosphere ($CO_2$, ~0.04%), oceans (largest exchangeable pool, as dissolved $CO_2$, $HCO_3^-$ and carbonate), fossil fuels, carbonate rocks, soil organic matter and living biomass.
- **Photosynthesis** removes $CO_2$ and fixes it into glucose; **respiration** by plants, animals and microbes returns it; **decomposition** of litter releases it.
- **Ocean exchange:** $CO_2$ dissolves at the surface and is used by phytoplankton for shells; some sinks as carbonate sediment.
- **Combustion** of fossil fuels, cement manufacture and land-use change release geologically stored carbon — this is the **fast** cycle overwhelming the slow one.
- **Human impact:** atmospheric $CO_2$ is nearly 52% above pre-industrial level, forcing warming and ocean acidification (lower pH), which harms coral and shellfish.

## L13. Describe the oxygen cycle.

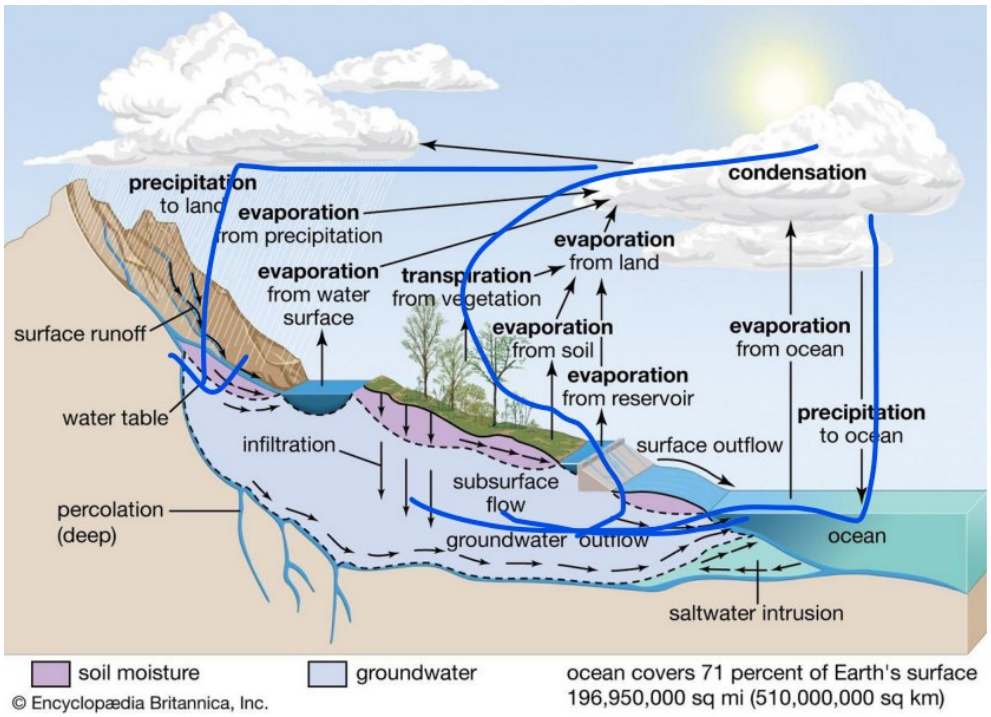
**7 marks**

The **oxygen cycle** is the movement of oxygen between its main reservoir, the atmosphere (21% $O_2$), and the biosphere, hydrosphere and lithosphere.

- **Source: photosynthesis** by plants, algae and cyanobacteria splits water and releases $O_2$; a small amount comes from photolysis of water vapour in the upper atmosphere.
- **Sinks: respiration** by all aerobic organisms; **combustion** of fuels; **decomposition** of organic matter; **oxidation** of minerals and metals (rusting, weathering of rocks); and formation of oxides in soil.
- **Ozone link:** $O_2$ in the stratosphere forms ozone ($O_3$) by photodissociation; the ozone layer absorbs UV and protects life.
- **Balance:** production and consumption are nearly balanced; burning fossil fuels and deforestation tilt the balance and lower dissolved oxygen in water bodies, harming aquatic life.
- The oxygen cycle is closely coupled with the carbon cycle — both are driven by photosynthesis and respiration.

## L14. Describe the hydrologic (water) cycle.

**7 marks**



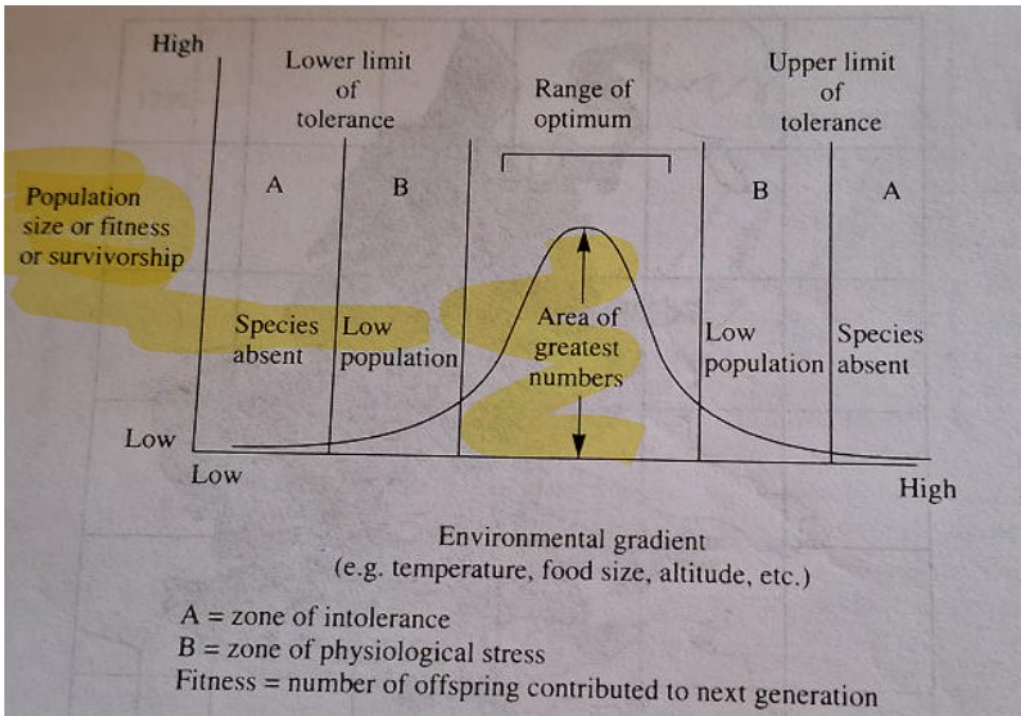

*The hydrologic (water) cycle — evaporation, condensation, precipitation, run-off and infiltration.*

The **hydrological cycle** is the continuous circulation of water between the oceans, atmosphere and land, driven by solar energy and gravity. Water is neither created nor destroyed, only moved and stored.

- **Evaporation** from oceans, lakes and rivers, and **transpiration** from plants (together, evapotranspiration) put water vapour into the air.
- **Condensation** of vapour on dust particles forms clouds and fog as air cools with altitude.
- **Precipitation** returns water as rain, snow, sleet or hail.
- **Infiltration and percolation** recharge soil moisture and groundwater aquifers; **run-off** flows over the surface into streams and rivers back to the sea.
- **Storage** occurs in oceans (97%), glaciers and ice caps, groundwater, lakes, soil and living organisms.
- **Importance:** supplies fresh water for drinking, irrigation and industry; distributes heat (latent heat); drives weathering and soil formation; sustains all ecosystems. Deforestation and urbanisation disturb infiltration and increase floods.

## L15. What is an environmental gradient? Draw and explain the tolerance curve, including under-population, optimum and over-population.

**7 marks**




*A schematic tolerance curve for a single species on one environmental gradient.*

An **environmental gradient** is a gradual change in an environmental factor (light, temperature, moisture, salinity, altitude) across space. Each species survives only within a range of that factor — its **limits of tolerance**.

- **Zone of intolerance (below minimum):** the factor is too low, the species cannot survive; population is zero — this is **under-population** caused by scarcity.
- **Zone of stress (low):** survival is possible but performance is poor; few individuals.
- **Optimum (zone of maximum):** the factor is most favourable, so abundance, growth and reproduction peak — the **optimal population**.
- **Zone of stress (high):** performance again falls as the factor becomes excessive.
- **Zone of intolerance (above maximum):** the factor is lethal — **over-population pressure** or overshoot beyond carrying capacity collapses the population (e.g. eutrophication killing a lake's fish, or heat stress in a crop).
- **Law of tolerance (Shelford):** the abundance of an organism is controlled by the factor for which it has the narrowest tolerance — the **limiting factor**. Species with wide tolerance (eury-) are widely distributed; narrow-tolerance (steno-) species are restricted.

## L16. Discuss Environmental Impact Assessment (EIA) and its steps.

**7 marks**

**EIA** is the formal process of identifying, predicting and evaluating the likely environmental effects of a proposed project before it is approved, and of proposing mitigation — with public participation. Its output is the **Environmental Impact Statement (EIS)**.

- **1. Screening:** decide whether the project needs a full EIA (size, location, sensitivity) and which category it falls in.
- **2. Scoping:** identify the key issues, boundaries and alternatives to be studied, in consultation with stakeholders.
- **3. Baseline study:** document the existing physical, biological and socio-economic environment.

- **4. Impact prediction and assessment:** estimate magnitude and significance of impacts on air, water, soil, noise, ecology and people.
- **5. Mitigation and the Environmental Management Plan (EMP):** measures to avoid, reduce or compensate impacts, with costs and monitoring schedules.
- **6. Public consultation/hearing and reporting** of the draft EIS.
- **7. Decision (appraisal and clearance), then monitoring and environmental audit** during construction and operation.
- **Importance:** prevents irreversible damage, saves cost by early identification, and makes development transparent and sustainable. **Limitation:** quality depends on data and on honest enforcement.

## L17. Discuss noise pollution — its causes, types of noise levels and effects on health.

**7 marks**




*Typical loudness of common sounds in dB, from breathing (0 dB) to a rocket launch (140+ dB).*

**Noise pollution** is unwanted or excessive sound that interferes with communication, rest and health. It is measured in **decibels (dB)**, weighted as dB(A) to match human hearing.

- **Causes/sources:** transport (road, rail, air traffic); industry (machines, compressors, generators); construction and mining (blasting, drilling); public address systems, loudspeakers and celebrations; domestic appliances; and neighbourhood activity.
- **Types by character: continuous** (fan, traffic hum), **intermittent** (aircraft, trains), **impulsive** (hammer, blast, gunshot). By source: occupational, transport, neighbourhood and domestic.
- **Levels:** 0–30 dB very quiet (breathing, whisper); 30–60 dB normal conversation; 60–85 dB busy traffic, factory (annoyance begins); 85–110 dB loud music, motorcycle, drilling (hearing risk with prolonged exposure); 120–140 dB jet aircraft, thunderclap (pain threshold ~120 dB); above 140 dB immediate damage.
- **Auditory effects:** temporary threshold shift, **noise-induced hearing loss** (permanent), tinnitus, and rupture of the eardrum at very high levels.
- **Non-auditory effects:** annoyance and irritability, sleep disturbance, hypertension and raised heart rate, stress and fatigue, reduced concentration and productivity, and impaired communication — hearing impairment is the most serious.

## L18. Describe the techniques of noise control and noise management.

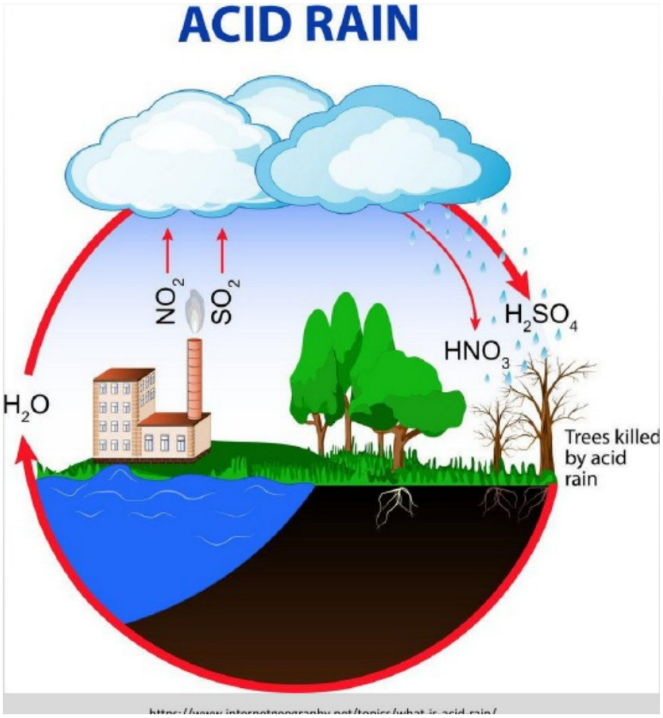
**7 marks**

Noise is controlled at three points: **at the source, along the path, and at the receiver** — in that order of preference.

- **1. At the source (most effective):** quieter machines and maintenance; mufflers and silencers on engines and exhausts; vibration isolators, dampers and resilient mountings; lubrication and tightening of parts; enclosing machines in acoustic hoods; replacing impact processes with pressing or welding.
- **2. Along the transmission path: noise barriers** and walls along highways; partial or full **enclosures**; sound-absorbing materials (glass wool, foam, perforated panels) on ceilings and walls; acoustic lagging of ducts and pipes; increasing distance between source and receiver; **green belts** of dense trees and shrubs.
- **3. At the receiver:** ear plugs and ear muffs (25–35 dB attenuation); job rotation to limit exposure time; periodic audiometry for workers; acoustic treatment of rooms; double-glazed windows for residences.
- **4. Administrative and legal:** zoning (keeping industry away from residential and silence zones); **Noise Pollution (Regulation and Control) Rules, 2000** with ambient limits and restrictions on loudspeakers and horns; time restrictions on construction and night-time activity; vehicle noise certification; public awareness and signboards near hospitals and schools.

## L19. What is acid deposition? Explain its mechanism and effects.

**7 marks**




*Formation cycle of acid rain — $SO_2$ and $NO_2$ react with atmospheric moisture to form $H_2SO_4$ and $HNO_3$, which fall and damage vegetation.*

**Acid deposition (acid rain)** is the fall of acidic components from the atmosphere — in rain, snow, fog (**wet deposition**) or as dry particles and gases (**dry deposition**). Natural rain is mildly acidic (pH ~5.6); acid rain is **pH below 5.6**.

- **Mechanism:** burning fossil fuels releases $SO_2$ and NOx. In the atmosphere they are oxidised:

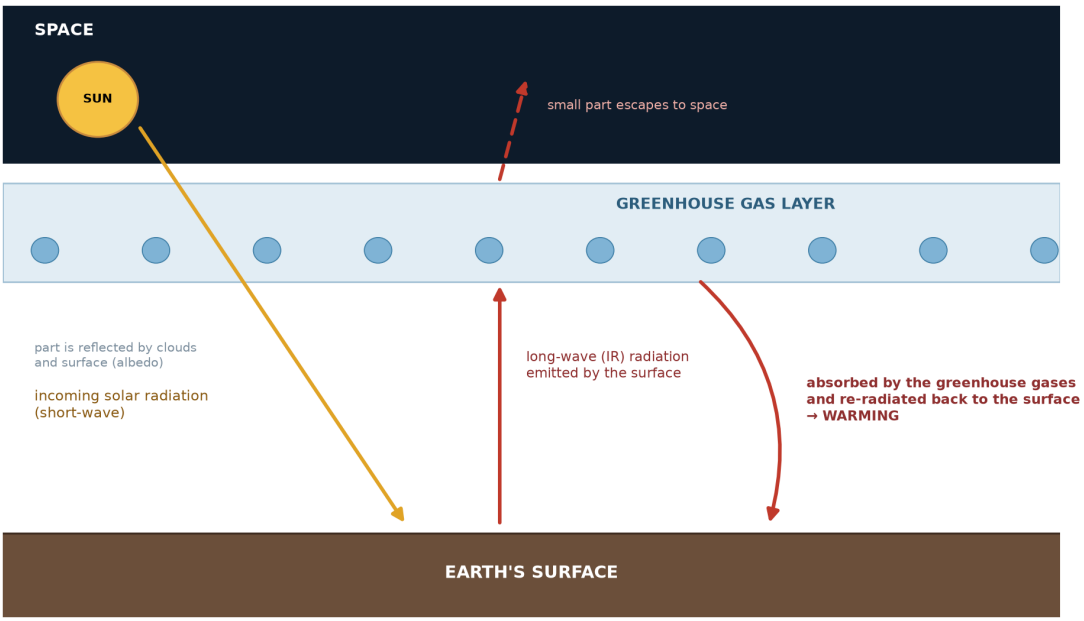
  $SO_2 + ½O_2 \rightarrow SO_3$ ; $SO_3 + H_2O \rightarrow H_2SO_4$ (sulphuric acid)

  $2NO_2 + H_2O \rightarrow HNO_3 + HNO_2$ ; $NO + O_3 \rightarrow NO_2 + O_2$ (nitric acid)

- The acids dissolve in cloud droplets and fall, or travel long distances before depositing — hence acid rain is a **transboundary** problem.
- **Effects: soil** — leaching of bases (Ca, Mg) and mobilisation of toxic $Al^{3+}$, loss of fertility; **water bodies** — lakes and streams acidify, fish eggs and fry die (pH < 5); **forests** — leaf damage and dieback; **materials** — corrosion of metals and erosion of limestone and marble monuments (Taj Mahal); **human health** — respiratory irritation from sulphates and nitrates.
- **Control:** low-sulphur fuels; **flue gas desulphurisation** (lime/limestone scrubbing); low-NOx burners and catalytic converters; tall stacks are not a solution (they spread the problem); liming of affected lakes.

## L20. Explain the greenhouse effect and global warming.

**7 marks**



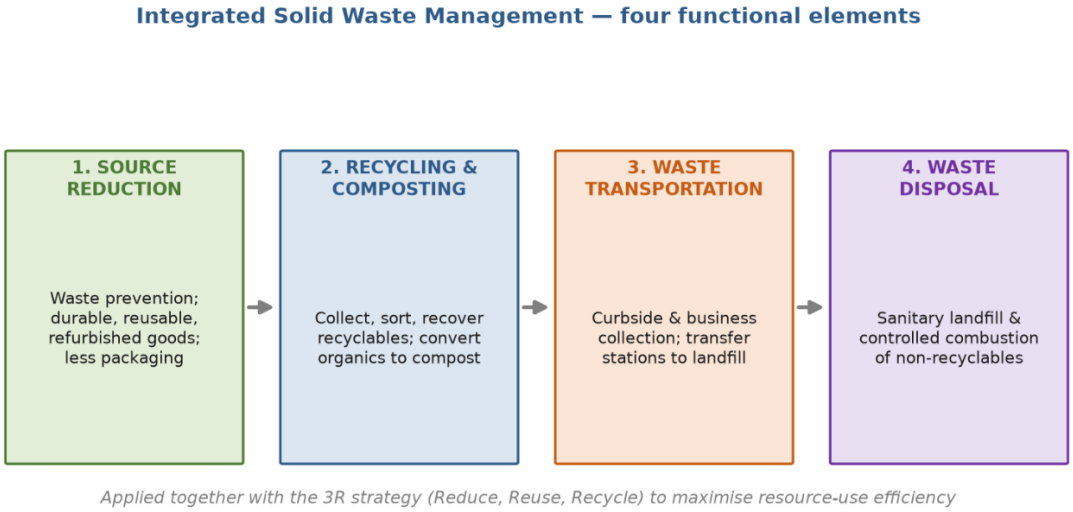

*The greenhouse effect — short-wave solar radiation enters, the surface re-emits long-wave IR, and greenhouse gases absorb and re-radiate much of it back, warming the surface.*

The **greenhouse effect** is the natural warming of the Earth's surface: the atmosphere is transparent to incoming short-wave solar radiation but absorbs the long-wave infrared radiation re-emitted by the warmed surface, and radiates much of it back. Without it the mean surface temperature would be about −18 °C instead of +15 °C.

- **Greenhouse gases:** $CO_2$ (largest share from fossil fuels), $CH_4$ (rice paddies, livestock, landfills; ~28× $CO_2$ per molecule), $N_2O$ (fertiliser, ~265×), CFCs (~10,000×, also ozone-depleting) and water vapour (the strongest natural one).
- **Enhanced greenhouse effect:** human emissions have raised $CO_2$ from ~280 ppm to over 420 ppm, thickening the layer, so less IR escapes and the energy balance shifts — the Earth warms. **Global warming** is this rise in mean surface temperature (~1.1–1.2 °C above pre-industrial).
- **Impacts:** melting glaciers and ice caps; sea-level rise and coastal flooding; more frequent heatwaves, cyclones and erratic monsoon; shifting crop belts and lower yields; coral bleaching and ocean acidification; spread of vector-borne disease; water scarcity and climate migration.
- **Mitigation:** shift to renewable energy and efficiency; reduce deforestation and plant trees; methane capture; **carbon capture and storage**; carbon pricing; and international agreements (UNFCCC, Kyoto, Paris). **Adaptation:** resilient crops, flood defences, early-warning systems.

## L21. What is Integrated Solid Waste Management (ISWM)? Explain its elements.

**7 marks**




*The four functional elements of integrated solid waste management.*

**ISWM** is a strategic, systems approach to managing solid waste — selecting and applying suitable technologies and management programmes to achieve waste-prevention, recycling and disposal goals while protecting public health and the environment.

- **1. Waste generation and characterisation:** quantifying and analysing waste (composition, moisture, calorific value) to choose the right system.
- **2. Collection, transfer and transport:** source segregation into wet/dry/hazardous, primary and secondary collection, transfer stations and route optimisation.
- **3. Recovery — recycling, composting and energy:** material recovery facilities, composting/vermicomposting of wet waste, RDF and biogas/incineration with energy recovery.
- **4. Disposal:** engineered sanitary landfill with liner, leachate collection and landfill-gas management — the last option, for residuals only.
- **Hierarchy followed:** Reduce > Reuse > Recycle > Recover (energy) > Dispose. **Stakeholders:** households, informal recyclers, municipalities, private operators, regulators, and producers under **EPR**.
- **Benefits:** less land required, lower greenhouse emissions, resource recovery, employment in the informal sector, and cleaner cities.

## L22. What is e-waste? Discuss its sources, composition and management.

**7 marks**



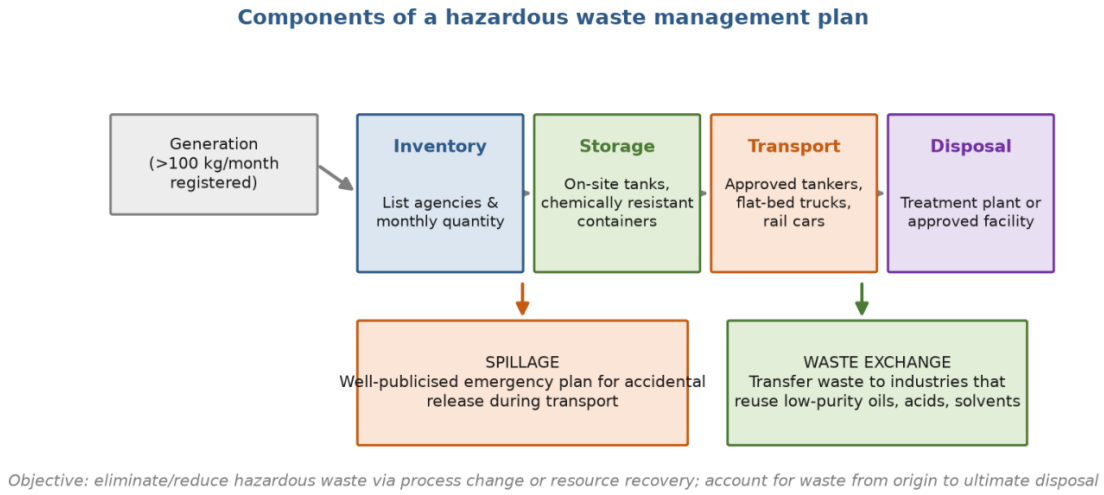

*Steps in e-waste recycling — collection, dismantling, shredding/separation and material recovery.*

**E-waste** is discarded electrical and electronic equipment — anything with a plug, battery or cord that has reached the end of its useful life. It is the fastest-growing waste stream because of short product cycles.

- **Sources:** households (TV, refrigerator, washing machine, mobile, laptop), IT and offices, telecom equipment, hospitals (diagnostic and monitoring devices), and manufacturing rejects.
- **Composition:** valuable metals (Cu, Al, Au, Ag, Pd), plastics, glass and ceramics — plus **hazardous** constituents: lead (CRT and solder), mercury (lamps), cadmium (batteries), hexavalent chromium, brominated flame retardants and PVC.
- **Management steps:** collection and take-back; manual **dismantling** to remove batteries, capacitors and toner; **shredding and separation** (magnetic, eddy-current, density); **recovery** of metals and plastics by hydrometallurgy/pyrometallurgy; treatment and secure disposal of hazardous residues.
- **Impacts if informal:** acid leaching and open burning release dioxins and heavy metals, contaminating soil, water and workers — including children.
- **Rules:** E-Waste (Management) Rules in India with **Extended Producer Responsibility**, authorised recyclers and the **Basel Convention** on transboundary movement of hazardous waste.

## L23. Define hazardous waste and discuss its classification and management.

**7 marks**



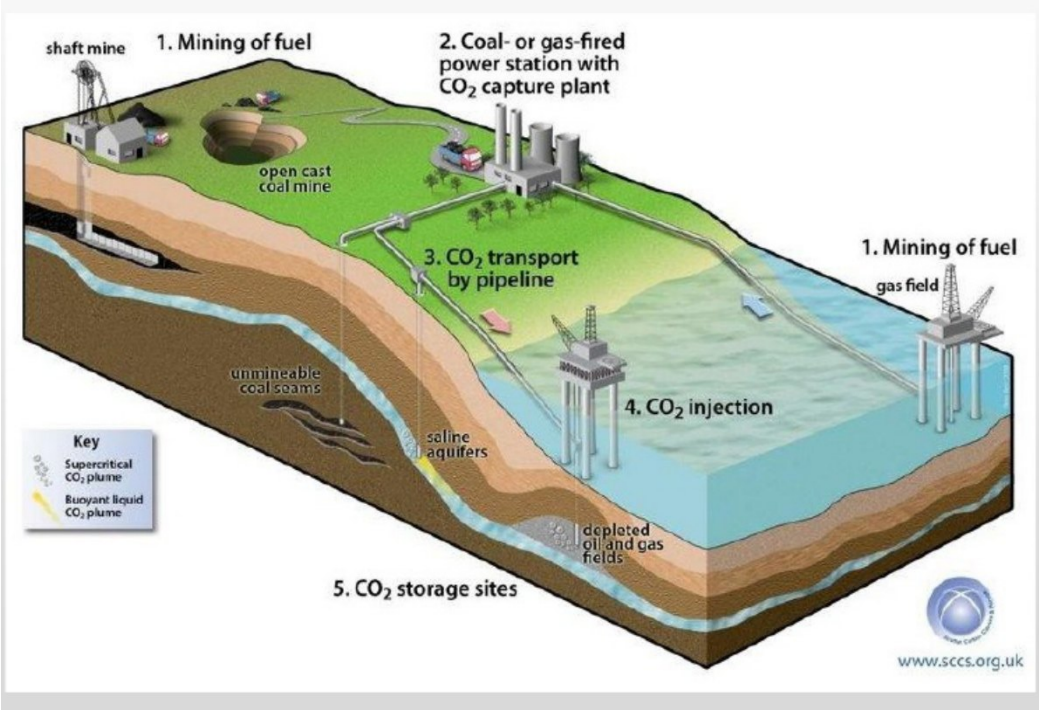

*Major components of a hazardous waste management plan.*

**Hazardous waste** is waste that, because of its quantity, concentration or physical, chemical or infectious character, may cause harm to human health or the environment. It is identified by four characteristics — **ignitability, corrosivity, reactivity and toxicity** (plus infectivity for biomedical waste).

- **Classification:** industrial (acids, alkalis, solvents, cyanides, heavy-metal sludge, spent catalysts); **biomedical/infectious** (hospital sharps, cultures, pathological waste); **chemical/pesticide** (obsolete stock, contaminated containers); **radioactive** (reactor and medical isotopes); and **e-waste**.
- **Management plan components:** (i) identification and inventory of wastes; (ii) segregation, labelling and compatible **storage** with bunded, ventilated areas; (iii) licensed **transport** with manifest/tracking; (iv) **treatment** — neutralisation, precipitation, incineration at high temperature, stabilisation/solidification, solvent recovery; (v) **secure landfill** with double liner and leachate collection; (vi) monitoring, record-keeping and reporting; (vii) contingency plans, training and medical surveillance.
- **Principles:** cradle-to-grave responsibility, polluter pays, proximity principle, and waste minimisation at source.

## L24. What is carbon sequestration? Explain the processes and techniques of carbon capture and storage (CCS).

**7 marks**



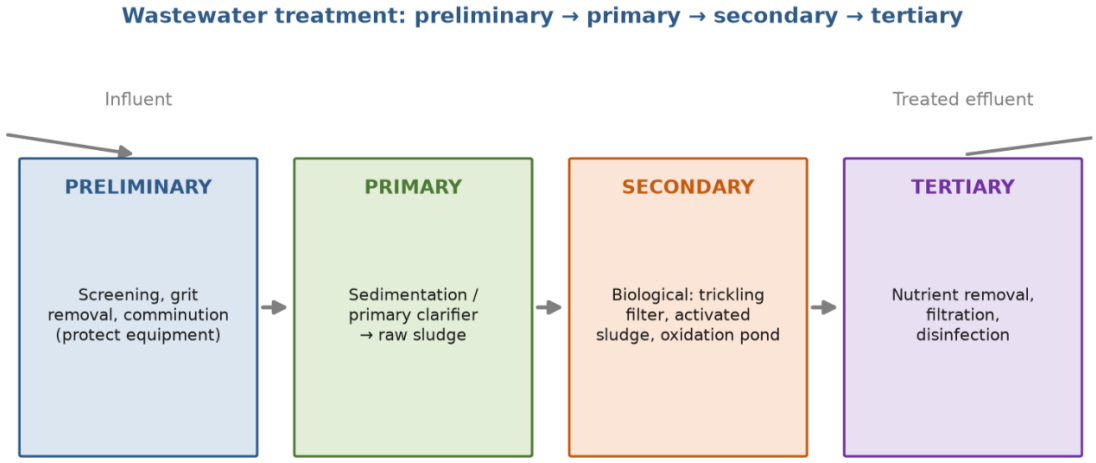

*Geosequestration — $CO_2$ captured from a power station is piped and injected into underground storage such as saline aquifers.*

**Carbon sequestration** is the capture and long-term storage of atmospheric $CO_2$ so that it does not contribute to the greenhouse effect. **CCS** is the engineered form: capture at a large point source, transport, and permanent storage.

- **Biological (natural) sequestration:** forests and afforestation (biomass carbon), soil organic carbon (conservation tillage, biochar), ocean uptake and algal photosynthesis, and wetlands.
- **Capture routes: post-combustion** (amine scrubbing of flue gas), **pre-combustion** (gasification to $H_2$ and $CO_2$), and **oxy-fuel combustion** (burning in pure $O_2$, giving a concentrated $CO_2$ stream).
- **Geological storage (geosequestration):** injection into deep saline aquifers, depleted oil and gas fields (with enhanced recovery), and unmineable coal seams; **mineral carbonation** fixes $CO_2$ permanently as carbonate rock.
- **Transport:** pipelines or ships to the storage site; **monitoring** for leakage and groundwater quality over decades.
- **Carbon credits:** a sequestered or avoided tonne of $CO_2$ becomes a tradable credit under cap-and-trade or offset schemes, which is how sequestration is monetised as policy.
- **Limitations:** high energy penalty and cost, storage-site availability, long-term leakage risk, and public acceptance.

## L25. Compare primary water treatment with tertiary (advanced) water treatment.

**7 marks**



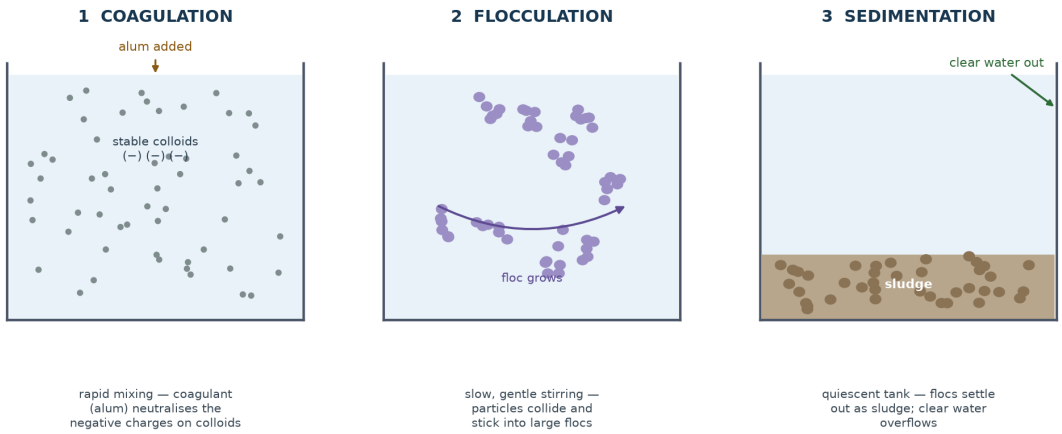

*The phases of wastewater treatment — preliminary, primary, secondary and tertiary.*

Wastewater treatment runs pre-treatment → primary → secondary → tertiary. **Primary** treatment is physical and removes settleable solids; **tertiary** treatment is advanced and removes what secondary treatment leaves behind, allowing reuse.

- **Primary:** screening and grit removal, then **sedimentation** in a primary clarifier (about 2 h detention) with sludge withdrawal. Removes ~60% of suspended solids and ~30–35% of BOD. Cheap, simple, no biological step.
- **Tertiary:** nutrient removal (nitrogen by nitrification–denitrification, phosphorus by chemical precipitation), **granular-media filtration**, **activated-carbon adsorption** for organics and colour, sometimes reverse osmosis or ion exchange, and **disinfection**. Removes >95% of solids, BOD, nutrients, pathogens and trace pollutants.
- **Cost and complexity:** primary is low-cost with low-skilled operation; tertiary is capital- and energy-intensive, needs chemicals and skilled operation.
- **Output:** primary effluent is still polluted (only fit for controlled discharge); tertiary effluent can meet strict river/irrigation standards and be reused for industry or, with further treatment, as potable water.
- **Placement:** primary is the first stage every plant has; tertiary is added only where the receiving water is sensitive (eutrophication, bathing, groundwater recharge).

## L26. Explain the mechanisms of coagulation, flocculation and sedimentation.

**7 marks**




*The three steps — coagulation (charge neutralisation, rapid mix), flocculation (slow stirring, floc growth) and sedimentation (flocs settle as sludge).*

These are the three successive steps of **chemical clarification**, used to remove the fine colloidal particles that settle too slowly on their own (clays, colour, micro-organisms).

- **1. Coagulation** — colloids carry negative surface charges and repel each other, so they stay in stable suspension. A coagulant (**alum, ferric chloride, ferrous sulphate, polyelectrolyte**) is added with **rapid mixing** (high velocity gradient, ~1 min). It hydrolyses to positively charged species that **neutralise the charge** (destabilisation) and form a sweep-floc network that enmeshes particles.
- **2. Flocculation** — gentle, slow stirring (10–30 min, low velocity gradient) makes the destabilised particles collide and stick together into visible **flocs**; polymers act as bridges between particles.
- **3. Sedimentation** — in a quiescent clarifier the flocs settle under gravity (Type-II settling) and are drawn off as sludge, while clear water overflows.
- **Controlling factors:** coagulant dose and type, **pH** (alum works best at pH 6–7.5), alkalinity, mixing intensity and time, temperature, and particle concentration.
- **Application:** the standard first stage of surface-water treatment for drinking water, and used in industrial effluent and tertiary phosphorus removal.

## L27. Discuss disinfection and sterilisation techniques used in water treatment.

**7 marks**

**Disinfection** destroys pathogenic organisms (but not necessarily spores) and is the final barrier in water treatment. **Sterilisation** destroys **all** micro-organisms including spores and viruses.

- **Chlorination** — the commonest method: $Cl_2$ or bleaching powder forms HOCl, which oxidises cell material. Cheap, gives a **residual** protection in the distribution system; disadvantages are taste/odour, trihalomethane formation and ineffective against cysts (*Giardia*, *Cryptosporidium*) at normal dose.
- **Ozonation** — very powerful oxidant, kills viruses and cysts, improves taste and colour; but no residual, so it is costly and needs energy.
- **UV radiation** (254 nm) — damages microbial DNA; effective against cysts and viruses, no chemicals or taste; needs clear water (low turbidity) and gives no residual.
- **Chlorine dioxide and chloramines** — used where THM formation must be avoided.
- **Boiling** — reliable household sterilisation (rolling boil for 1–5 min).
- **Membrane filtration** — MF/UF remove protozoa and bacteria; RO removes viruses too.

- **Sterilisation methods** (for media and equipment): **autoclave** — 121 °C at 15 psi for 15–20 min; **dry heat** — 160–170 °C for 2 h; **ethylene oxide** gas for heat-sensitive items; **gamma irradiation**; and membrane filtration for liquids.
- **Efficiency depends on:** contact time, dose/concentration, pH, temperature, turbidity and the type and number of organisms present.

# PART C
# Numericals — formulas with one worked line each

The briefing expects 2–3 numericals in the short section and a few in the long section; these are the six types it names.

### N1. Net primary productivity

**formula + example**

**NPP = GPP − R**, where GPP is gross primary productivity and R is the respiration loss of the producers.

If GPP = 1200 kcal $m^{-2}$ $yr^{-1}$ and R = 400 kcal $m^{-2}$ $yr^{-1}$, then NPP = 1200 − 400 = **800 kcal $m^{-2}$ $yr^{-1}$** — this is the energy available to herbivores.

### N2. 10% law across trophic levels

**formula + example**

Energy transferred to the next level = 10% of the energy of the present level.

If $T_1$ = 10,000 kcal, then $T_2$ = 1,000, $T_3$ = 100, $T_4$ = 10 kcal. Energy lost to the atmosphere between $T_1$ and $T_4$ = 10,000 − 10 = **9,990 kcal (99.9%)**.

### N3. Sound power level (L<sub>W</sub>)

**formula + example**

**$L_W = 10 \log_{10} (W / W_0)$** dB, where $W_0 = 10^{-12}$ W.

For W = $10^{-3}$ W: $L_W = 10 \log_{10} (10^{-3}/10^{-12}) = 10 \log_{10} 10^9$ = **90 dB**.

### N4. Sound pressure level (L<sub>P</sub>)

**formula + example**

**$L_P = 20 \log_{10} (P / P_0)$** dB, where $P_0$ = 20 μPa (reference pressure).

For P = 30 μPa: $L_P = 20 \log_{10} (30/20) = 20 \times 0.176$ = **3.5 dB**. For P = 20 × $10^3$ μPa: $L_P = 20 \log_{10} (20{,}000/20) = 20 \times 3$ = **60 dB**.

### N5. Vehicle and population emissions

**formula + example**

Multiply the per-vehicle (or per-person) emission by the number of vehicles (or people) and by the number of days to get the yearly load.

A car emits 0.25 kg CO per day. For 5,000 cars over one year: 0.25 × 5,000 × 365 = **456,250 kg ≈ 456 t of CO per year**.

## N6. BOD and its modified forms

**formula + example**

**$BOD_5 = (D_1 - D_2) \times$ dilution factor**, i.e. BOD = p ($DO_i - DO_f$) with p the dilution factor; for an undiluted sample p = 1.

> Sample diluted 100 times; DO falls from 9.0 to 2.0 mg/L: $BOD_5 = (9.0 - 2.0) \times 100 =$ **700 mg/L**.

- **Seeded BOD:** subtract the seed correction — BOD = $[(D_1 - D_2) - (B_1 - B_2)f] \times p$, where $B_1$, $B_2$ are the seed-blank DO values and f the seed fraction.
- **Ultimate BOD:** $BOD_t = L_0(1 - e^{-kt})$; at 20 °C, k ≈ 0.1 $day^{-1}$ for domestic sewage, and $BOD_5 \approx 0.68\ L_0$.
- **COD** is measured chemically ($K_2Cr_2O_7$ digestion) and is always higher than BOD; the BOD/COD ratio indicates biodegradability (>0.4 = easily biodegradable).

**Source note.** Question topics are taken from the Gemini analysis of the class video (the meeting summary); answers are written to the depth of the CY202 syllabus and the model answers in the five unit question banks, with the five topics that the banks do not cover — noise standards, metal centres, coagulation–flocculation–sedimentation, disinfection–sterilisation and vehicle emissions — written out in full here.